

Instructions

Allowed tools: Paper, pen, five pages of notes and dictionary.

Write your name and personal number here:

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Write the secret number given to you on EVERY page and here:

.....

If you want your result sent to you by email, write your email address here:

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This is the exam for this course, and it will contribute to the final grade of 50%; 30% of the final grade will be based on the project, 10% on the lab queries, and 10% on discussion/pre-lecture queries. To pass the course, you need to pass all labs with at least 50% and get at least 40% on this exam.

Minimum score per letter grade (truncated, not rounded)

- A: 92%
- B: 84%
- C: 76%
- D: 68%
- E: 60%
- Fx: 55%
- F: <55%

Each question should be answered within the provided space. If you need more space, write it on the back of the paper.

You can answer in either English or Swedish.

Arne can be reached at 070-6951045 for questions and will call in a few times during the exam. Petras and Samuel will come by to take questions.

Formulas that you might find helpful (from wikipedia):

$$H = - \sum_i p_i \log_2(p_i)$$

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x) p(y)},$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

$$S_{ij} = \frac{1}{\lambda} \log \frac{p_{ij}}{q_i q_j}$$

In computing S_{ij} , we will take $\lambda=1$.

q_i is the marginal probability of occurrence of amino acid of type i .

p_{ij} is the joint probability of the pair of amino acid types i and j .

$\log$ is base-2.

Question 1: Substitution matrices (10 p):

Substitution matrices

- Describe how PAM matrices are constructed.
- Describe how BLOSUM matrices are constructed.
- Analyse the similarities and differences between them.

Question 2: Profile Hidden Markov Model (10 p)

Construct a profile Hidden Markov Model for the multiple sequence alignment below.
To get full points, you need to:

- Define the model states and draw a diagram showing the profile HMM
- Calculate non-zero probabilities
- Show the path of sequence No. 3 through the profile HMM.

Here is the Multiple sequence alignment:

Sequence 1	GDYYW---KLY
Sequence 2	GDY-WHQQKLP
Sequence 3	--YYAHQQKLP
Sequence 4	GDYYA---RIP
Sequence 5	GEY-A---RLW
Sequence 6	AEYFA---KIW
Sequence 7	GEFFAH--KIW
Sequence 8	GEYYA---KLY
Sequence 9	ADYYW---RIP
Sequence 10	-EYY----RIP

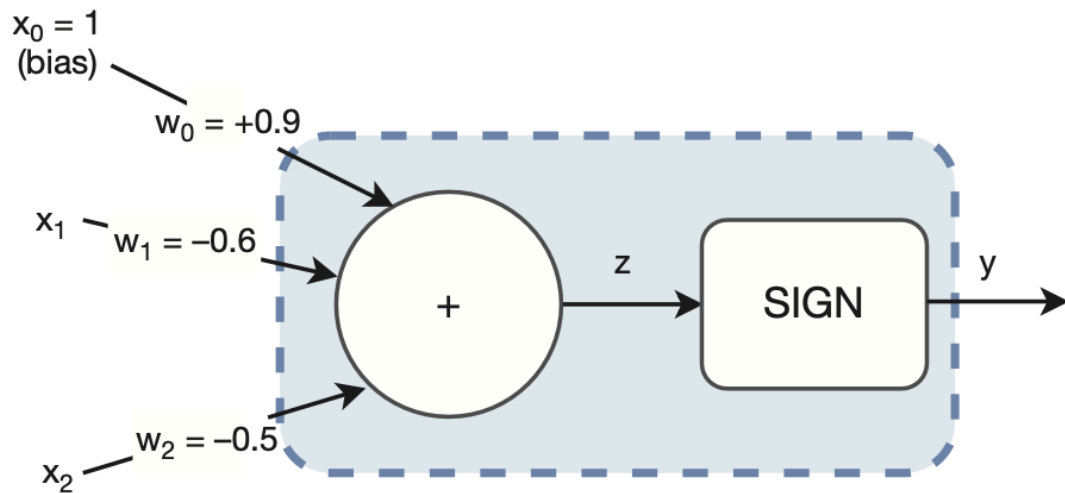
Question 3: Comparison of protein structures (10 p):

- Describe what metrics can be used to compare tertiary structures.
- Describe what metrics can be used to compare quaternary structures.
- Analyse the similarities and differences between those two groups of metrics.

Question 4.

Ekman Ch 1: the Rosenblatt perceptron.

Consider the following perceptron:

Draw the output signal in the X_1 - X_2 plane.

2 pt: Draw a line separating the X_1 and X_2 regions that returns True (+1) from that which returns False (-1).

2 pt: What is the X_1 - intercept?

2 pt: What is the X_2 intercept?

1 pt: On your drawing above, mark the side of the line that returns +1.

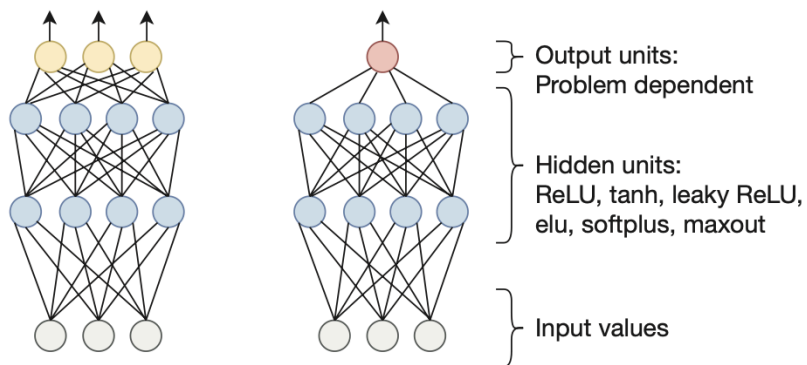
1 pt: Mark the side of the line that returns -1.

2 pt: What logical operator is this?

Question 5

Ekman Ch 6

Consider the following networks:



2 pts: Imagine you wish to solve a regression problem, for example, by predicting the housing price based on information about the property (e.g., district, crime rate, number of rooms). What activation function should you use on the output layer?

2 pts: Still regarding regression, what loss function should you use?

1 pt: Which of the networks looks most like it would work for multiclass classification?

2 pts: Still, regarding multiclass classification, what sort of activation function should you use in the output layer?

1 pt: Which one could be used for binary classification (assuming an appropriate choice of activation functions)?

2 pts: Still regarding binary classification, what activation function should you use in the output layer?

Question 6

Ekman Ch 8

Consider the Inception module (Szegedy, Liu, et al., 2014) shown below.

The blue layer (4 channels) is the input layer.

Note that there are four parallel streams taking input from the input layer, as follows:

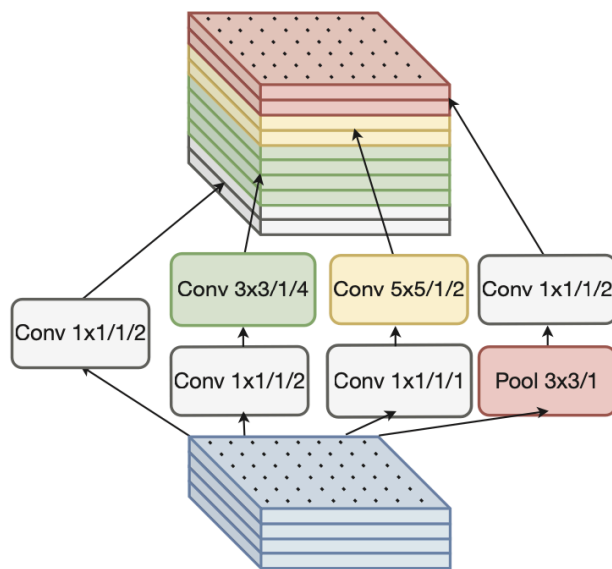
Stream 1: Convolution with 1x1 kernel, 1x1 stride, and 2 output channels.

Stream 2: Convolution, 1x1 kernel, 1x1 stride, 2 output channels. Followed by convolution, 3x3 kernel, 1x1 stride, 4 output channels.

Stream 3: To the right of stream 2.

Stream 4: Max pooling, 3x3 kernel, stride of 1. Followed by Convolution, 1x1 kernel, 1x1 stride, 2 channels.

The output consists of channels that are concatenated from the output of the four streams.



2 pts: What is the point of Stream 1? Name all that are reasonable and possible.

2 pts: Regarding Stream 2. Why have the 3x3 convolution after a 1x1 convolution? Why not have the 3x3 convolution read input directly from the input layer?

2 pts: Compare Streams 2 and 3. Stream 2 begins with a 1x1 kernel and two output channels. Stream 3 begins with a 1x1 kernel and one output channel. Why the difference in output channels?

2 pts: Regarding Stream 4. Why do we NOT have a 1x1 convolutional layer BEFORE (meaning, closer to the input) the 3x3 max pooling layer?

2 pts: Still on Stream 4. What is the point of the 1x1 convolutional layer that appears after the 3x3 max pooling layer?

Question 7: LLMs (10p)

Large language models (LLMs) are used to solve many problems in life science. A key concept in LLM is attention, which is based on optimising three matrices: $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$.

- Describe how the Query (**Q**), Key (**K**) and Values (**V**) are obtained from these matrices.
- Describe how these are used to obtain the attention value, i.e. the function $f(Q,K,V)$ and how the vectors Q, K, and V are obtained from an input vector (embedding) X, i.e., what is $f()$ in this equation
 - $\text{Attention}(Q,K,V) = f(Q,K,V)$

Question No 8: AlphaFold

AlphaFold2 was a breakthrough in protein structure prediction. It used several tricks to achieve such good performance. In particular, the architecture was designed ingeniously. To describe that you understand this, you should answer these two questions.

- Draw the architecture of AlphaFold2 - starting with the input and ending with the final output. Each computational block only needs to be described briefly, but all connections are important, details of different modulus are not necessary to be included (5p).
- List and explain the three main differences between AlphaFold2 and AlphaFold3 architectures (5p).